HumanPulse Protocol

Privacy-first Proof of Unique Humanity infrastructure for Solana.

HumanPulse proves the presence of a unique living human at a specific moment in time — instantly, privately, and on-chain.

MVP focus: Sybil resistance for DAOs, airdrops, and token launches

Current implementation: webcam-based liveness, challenge-response, Solana Devnet token, and Anchor burn-on-use

Positioning: protocol-grade uniqueness layer, not a centralized KYC provider

Hackathon MVP • Solana Devnet

Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

PROBLEM

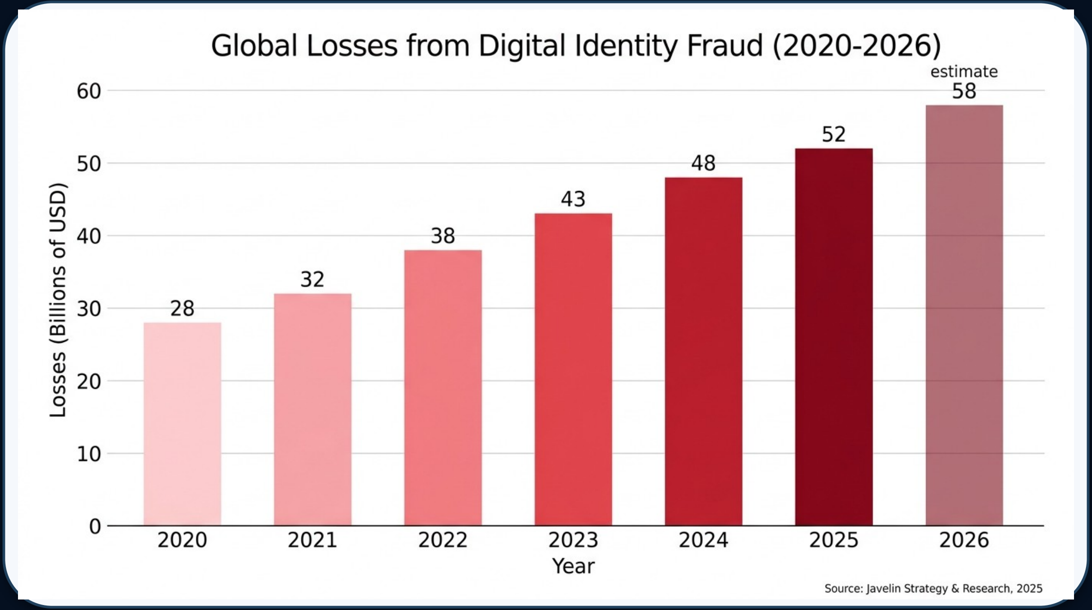
The internet cannot tell humans from machines

This creates a growing attack surface for airdrops, DAO governance, token launches, and identity-sensitive consumer flows.

$52B
digital identity fraud losses in 2025

800K+
Sybil wallets identified in one large airdrop event

Today
CAPTCHAs, passwords, and weak liveness are breaking down



Centralized identity

Introduces custody, friction, and privacy risk.

Weak anti-bot tools

Generic CAPTCHAs and static checks are increasingly bypassable.

Missing primitive

Crypto lacks a neutral, privacy-aware proof of human presence.

What HumanPulse is designed to do

Significantly increase the cost of Sybil attacks while minimizing the amount of personal data required.

Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

SOLUTION

What HumanPulse proves in one verification

The protocol combines live presence checks, an unpredictable challenge, and on-chain proof consumption in a low-friction flow.

1. Live human

Webcam-based liveness checks that a live person is present now.

2. Challenge

The user responds to a real-time unpredictable prompt.

3. Proof consumed

The result is turned into a proof interaction that can be consumed on-chain.

4. Use on Solana

An application gates access, voting, or eligibility using the proof.

No biometric database

HumanPulse is intended to avoid storing raw biometric data on-chain or creating a centralized biometric vault.

Privacy-first UX

The complexity of proof generation is hidden from the user; the goal is fast verification with minimal disclosure.

Built for crypto use cases

Primary MVP use cases are DAO voting, airdrop eligibility, anti-bot gating, and token launch protection.

Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

CURRENT MVP

What we built for the hackathon

The current implementation demonstrates a practical HumanPulse flow on Solana Devnet. It is an MVP, not a fully production-hardened identity network.

Webcam-based liveness

Current MVP verification begins with live webcam input.

Challenge-response

An unpredictable prompt raises the attack cost for spoofing.

Solana Devnet $HPP token

Devnet token used inside the current proof consumption flow.

Anchor burn-on-use program

On-chain burn logic demonstrates consumption of a verification primitive.

Demo website / widget

End-to-end user flow packaged for hackathon demonstration.

Public artifacts

Technical whitepaper, demo video, figures, and pitch deck.

Scope discipline for judges

Current MVP / hackathon implementation:
- Sybil resistance for DAOs, airdrops, and token launches
- Webcam-based liveness
- Challenge-response verification
- Solana Devnet token and Anchor burn-on-use

Technical roadmap and optional research modules are described later and are not represented as fully implemented.

Hackathon deliverable

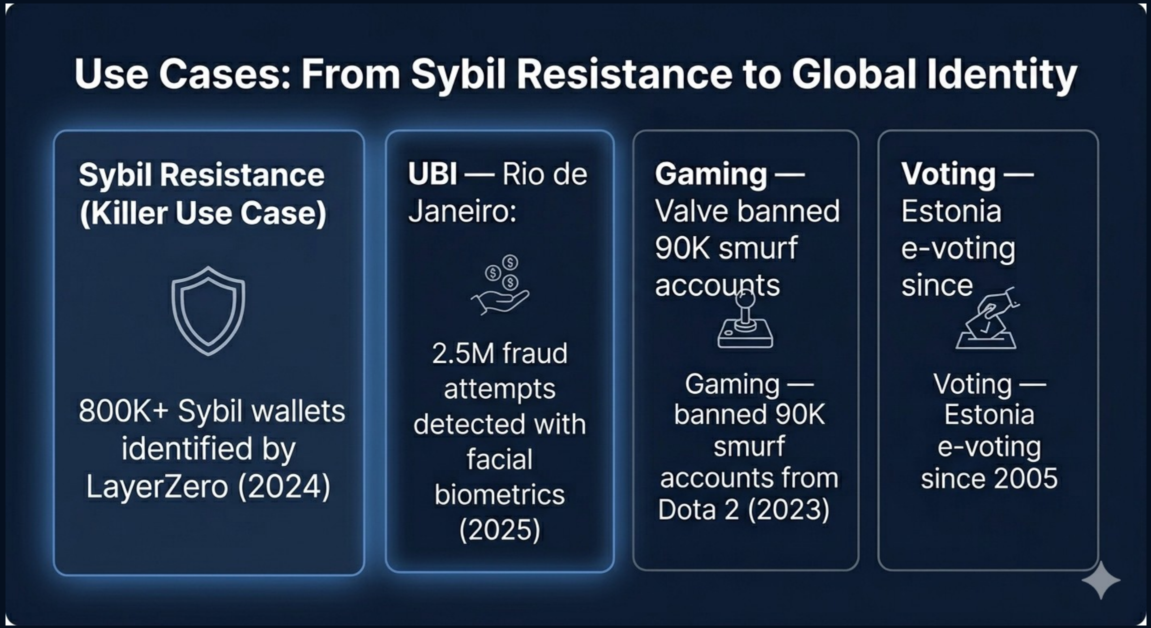
A working demonstration of proof-of-human gating on Solana Devnet, packaged with a live site, demo video, repository, and whitepaper.

Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

USE CASES

Start with the highest-urgency wedge: Sybil resistance

HumanPulse begins where the pain is immediate: incentive-heavy systems where fake or duplicated participation distorts outcomes.



Initial wedge

DAO voting, airdrops, token launches, and anti-bot gating are the clearest near-term applications.

Why it matters

A uniqueness layer can reduce false participation, improve fairness, and make incentive programs harder to game.

Longer-term direction

If the primitive is strong enough, the same network can support broader identity-sensitive consumer and governance flows.

Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

HumanPulse architecture: three layers, one claim

The protocol design combines liveness, challenge-response, and proof generation into a stacked uniqueness architecture.

1. Liveness detection

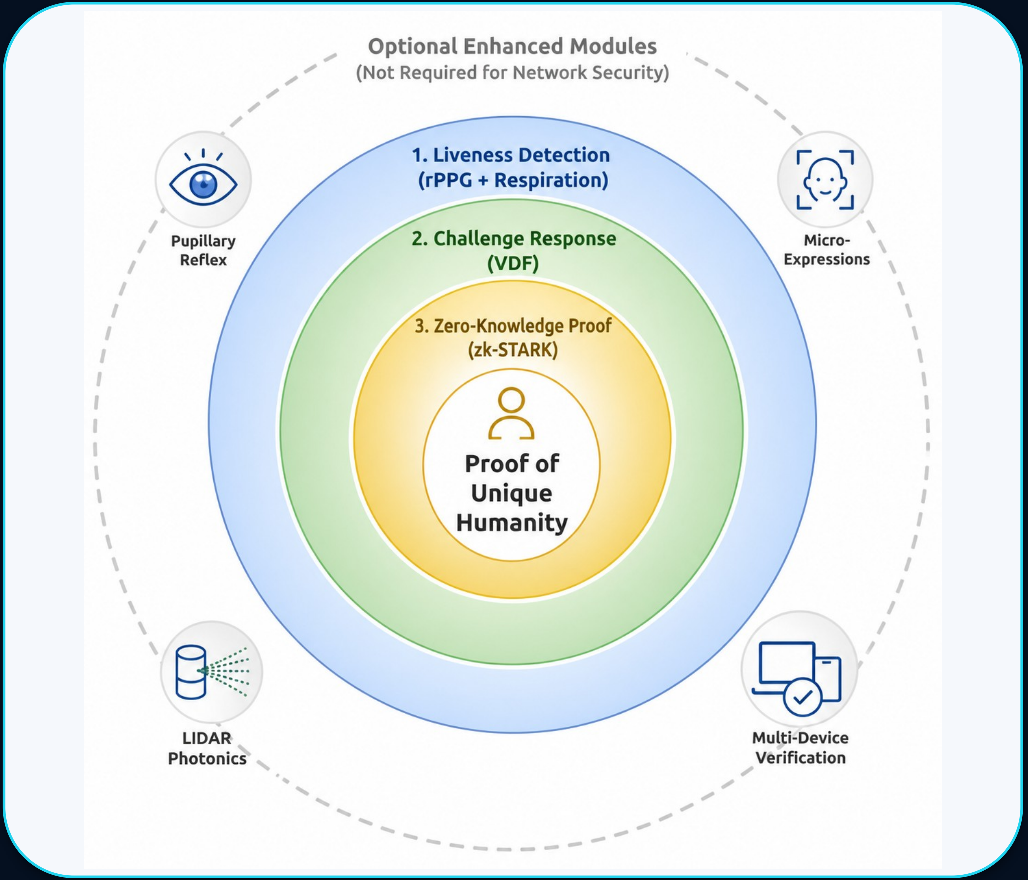
The protocol begins by measuring whether a live human is present. The MVP demonstrates webcam-based liveness; richer signal layers appear in the roadmap.

2. Challenge-response

A live challenge introduces unpredictability and makes replay or pre-rendered spoofing more difficult.

3. Proof layer

The whitepaper’s protocol direction culminates in zero-knowledge proof generation. The current MVP demonstrates the consumption side of the primitive on Solana Devnet.



Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

Privacy is an architectural constraint

HumanPulse is intended to prove human presence without turning the protocol into a biometric data honeypot.

No raw biometrics on-chain

On-chain logic consumes proof outcomes, not raw face data.

Temporary processing

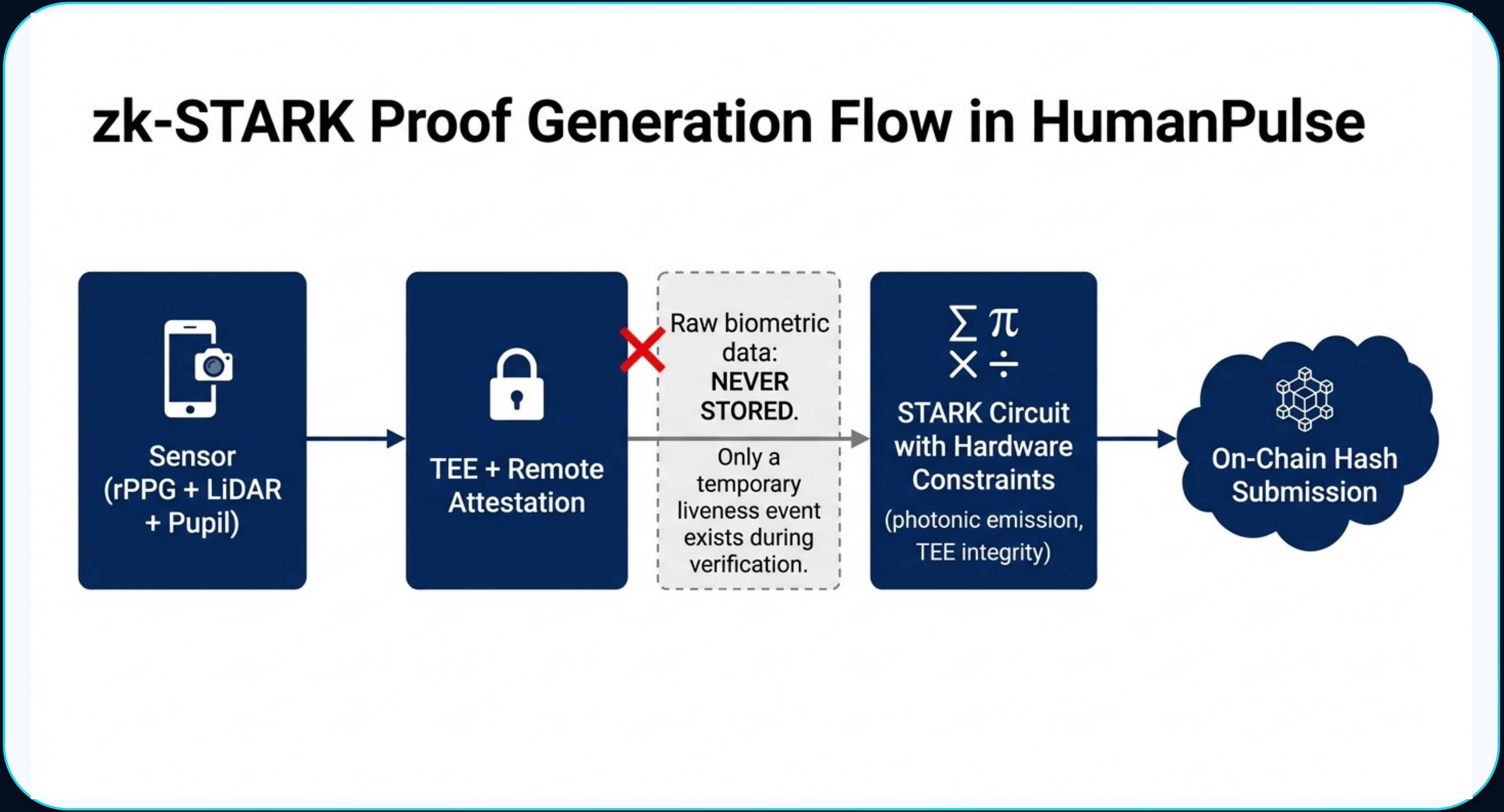
The protocol direction minimizes persistent exposure of sensitive inputs.

Proof, not identity disclosure

The target primitive is uniqueness verification, not full identity revelation.

Research roadmap

Recursive zk-STARK aggregation, TEEs, and local proof generation are roadmap modules.



Source: HumanPulse Technical Whitepaper v7.2. Research modules shown here are roadmap elements, not all current MVP features.

Why Solana is the right execution layer

The current MVP uses Solana Devnet and an Anchor program to show how a uniqueness primitive can be consumed in a fast, low-cost on-chain flow.

Low-cost interactions

Frequent verification usage requires inexpensive on-chain actions.

Fast finality

Consumer-facing gating flows benefit from quick confirmation.

Programmable token logic

The burn-on-use pattern is implemented via an Anchor program on Devnet.

Composable applications

DAO voting, access control, launches, and rewards can all plug into the same primitive.

Developer fit

Anchor and Solana tooling make it practical to build an MVP quickly and clearly.

Current on-chain artifacts

Devnet program: D9tRQi8n...Q5B5N
Devnet $HPP token: EsBZKfFm...ZSnR

MVP implementation

Current MVP: Solana Devnet token + Anchor burn-on-use program + demo website/widget. Technical roadmap: stronger proof aggregation, validator / aggregator architecture, and expanded privacy-preserving proof flows.

A usage-driven $HPP economy

The whitepaper frames $HPP as a network primitive tied to proof usage. The current MVP demonstrates the burn-on-use concept on Solana Devnet.

Proof consumption

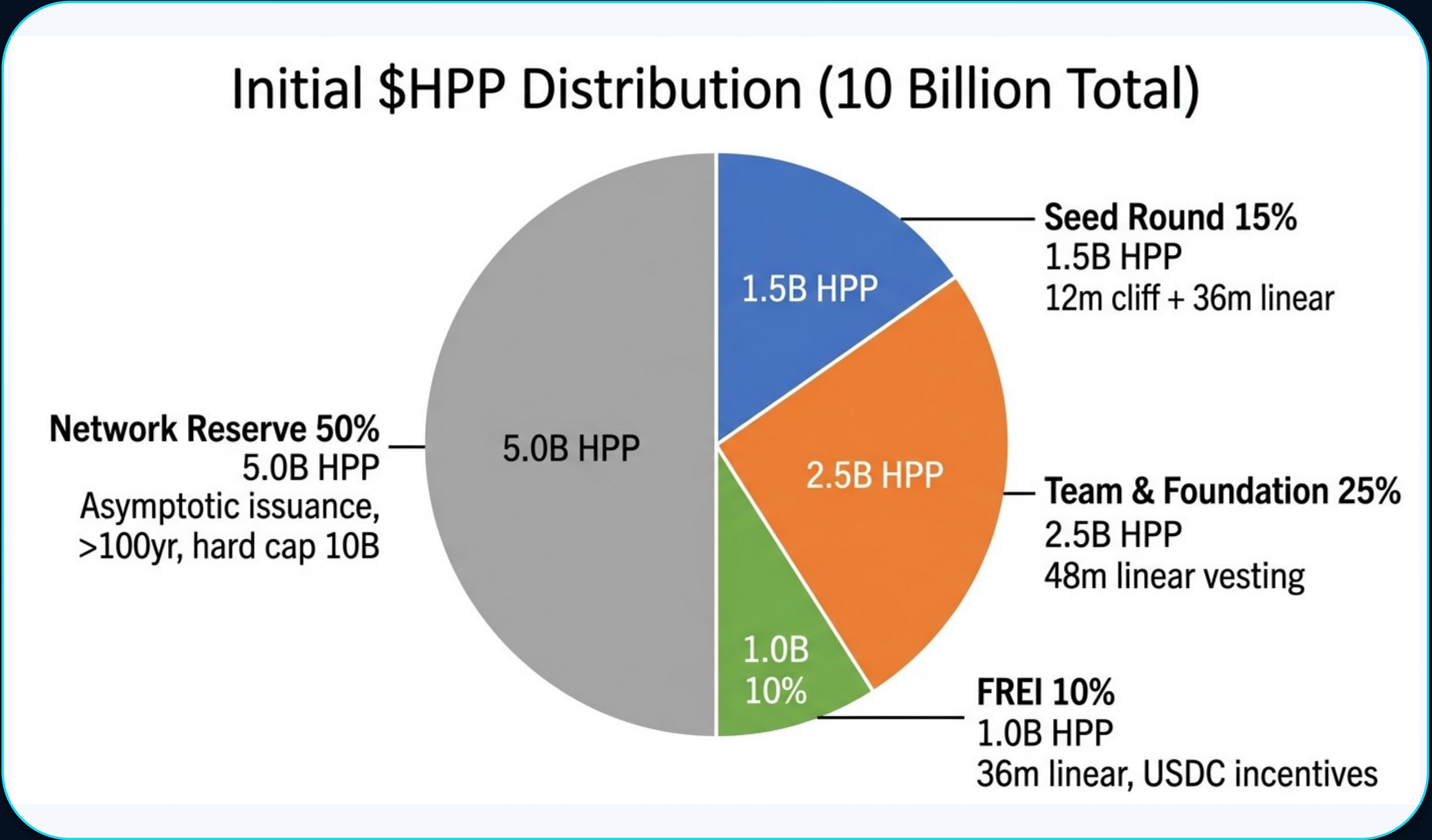
Verification is meant to be used, not merely held. Burn-on-use aligns token mechanics with activity.

MVP scope

The Devnet implementation demonstrates the mechanism; it is not presented as finalized production token economics.

Longer-term design

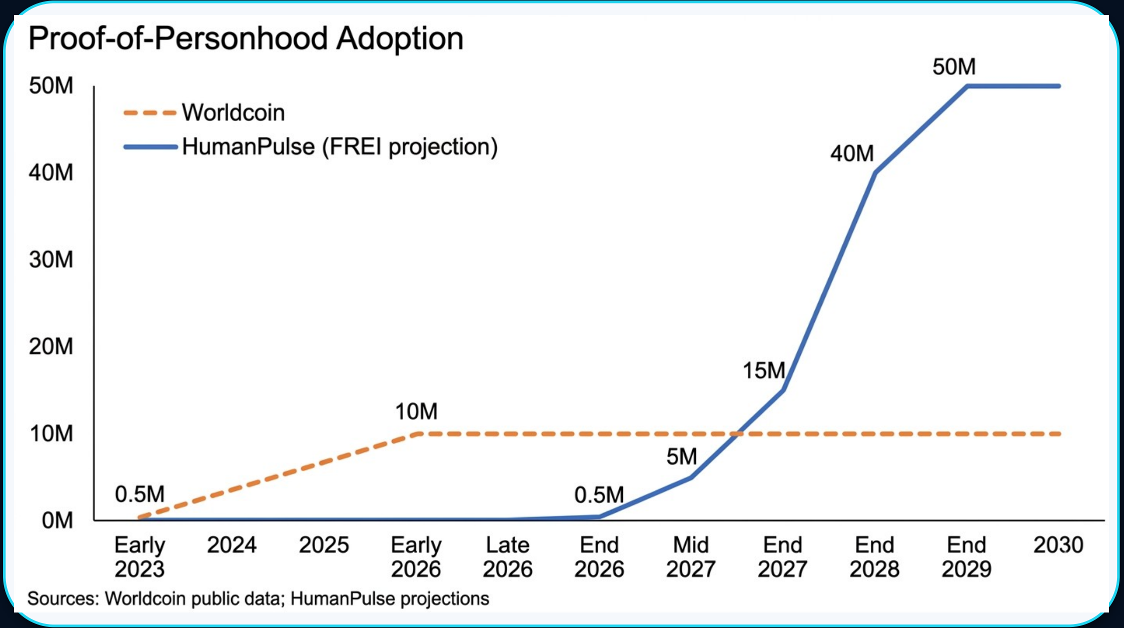
Broader token distribution, reserve structure, and network incentives follow governance and protocol maturation.

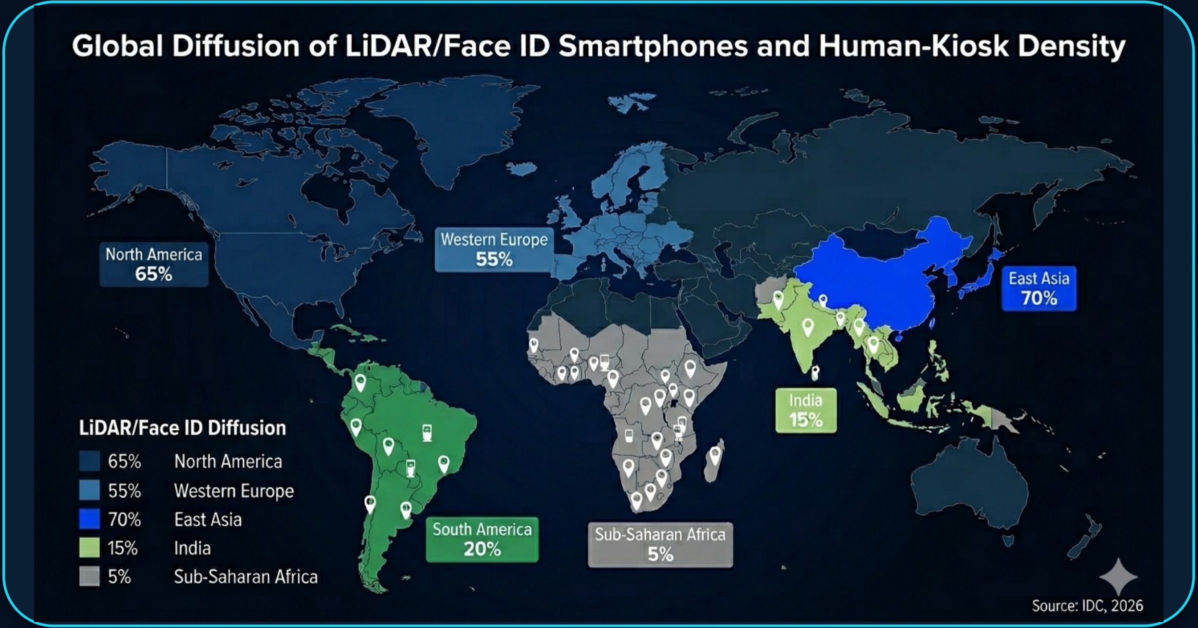


Source: HumanPulse Technical Whitepaper v7.2. Token structure shown is from the current protocol narrative and should be interpreted with roadmap context.

From wedge to infrastructure

HumanPulse starts with anti-Sybil applications, but the long-term opportunity is a neutral uniqueness layer that can compound across digital ecosystems.





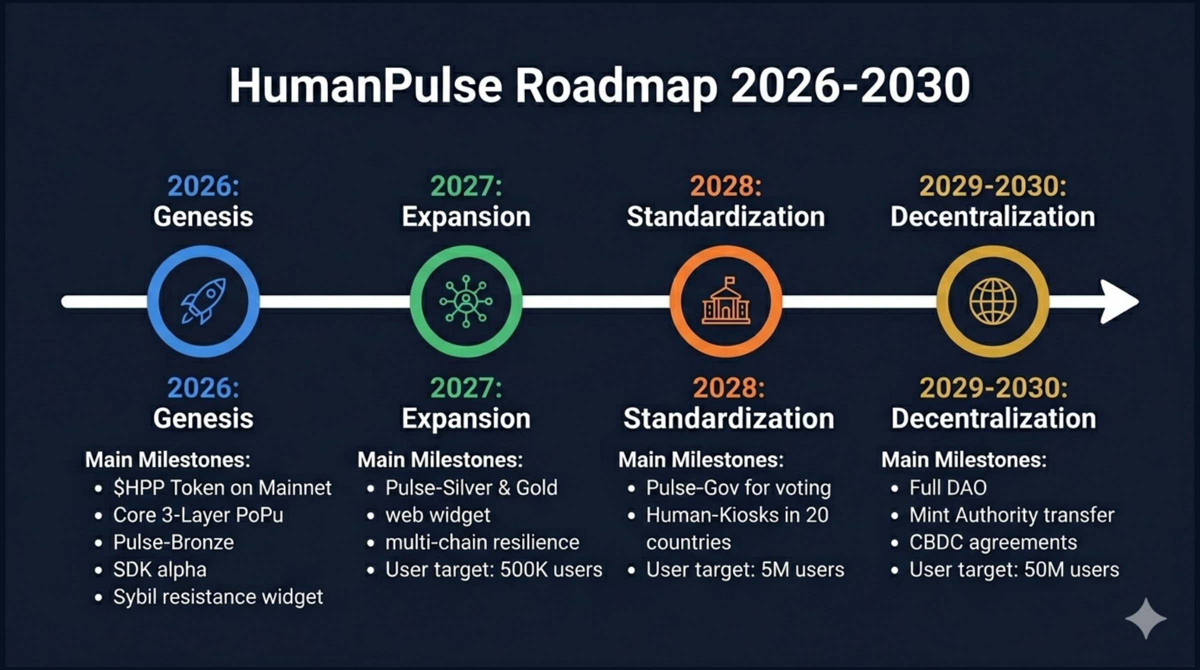
Near-term
Sybil-resistant DAO, airdrop, and token launch flows.

Mid-term
Wallets, exchanges, social applications, and consumer gating.

Long-term
A protocol-grade uniqueness layer for the AI era.

From hackathon MVP to protocol-grade infrastructure

The roadmap separates near-term protocol hardening from optional research modules that may increase robustness over time.



Near-term buildout

Validator / aggregator layer, stronger proof pipeline, deeper integrations, and production-oriented hardening.

Research modules

Recursive zk-STARK aggregation, TEE-based local proof generation, LiDAR photonics, pupillary reflex, micro-expression analysis, and multi-device verification.

Message to judges

The hackathon MVP demonstrates the direction clearly today. The roadmap shows how the protocol can evolve without overstating current implementation status.

Source: HumanPulse Technical Whitepaper v7.2 and current MVP implementation.

CLOSING

Prove humanity. Preserve privacy. Build on Solana.

HumanPulse is designed to provide a privacy-first proof of unique human presence for the AI era.

GitHub Repository

github.com/humanpulseprotocol/HumanPulse-Protocol

Live Website

humanpulse-landing.vercel.app

Demo Video

humanpulse-landing.vercel.app/HumanPulse-demo.mp4

Technical Whitepaper

github.com/humanpulseprotocol/HumanPulse-Protocol/blob/main/HumanPulse%20Protocol/whitepaper/HumanPulse_Technical_Whitepaper_v7.2_FINAL.pdf

Solana Devnet Program

explorer.solana.com/address/D9tRQi8nZzARTJZXTmSsy4hrx4BwKUNPezKzayvQ5B5N?cluster=devnet

$HPP Token on Devnet

explorer.solana.com/address/EsBZKfFmhVAypmVm6QmenevZ56XZQZUjYF8CLzVnZSnR?cluster=devnet

HumanPulse proves the presence of a unique living human at a specific moment in time — instantly, privately, and on-chain.